Lightweight Microchannel Thermal Management for Orbital AI Data Centers

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Introduction

With AI taking over much of Earth, orbital data centers have become increasingly appealing. They create their own energy via solar cells and cool themselves with recycled water (or some other coolant). However, working in a vacuum creates challenges unique to space. Namely: cooling. Without air, the only viable method of heat rejection is radiation, forcing a large redesign of current cooling networks. We have designed a radiator constructed of a plastic pipe array connected by an inlet and outlet manifold (a larger pipe carrying coolant to and from the radiator). We have focused on optimizing the pipe geometry, material properties, and fabrication processes, minimizing weight to reduce deployment costs.

Optimizing Geometry

Used COMSOL Multiphysics to simulate heat rejection:

- Model laminar fluid flow with heat transfer and radiation through three cooling pipes with a fixed fluid inlet temperature of $370K$
- Material properties of High-Density Polyethylene (HDPE)
- Calculate power dissipated and mass of one pipe
- Scatter power dissipated against pipe mass, calculate the Pareto Frontier
- Find the point that forms the steepest line through the origin

Acknowledgments

This project couldn't have been done without the help of Professor Igor Bargatin and PhD candidate Oreva Omodior, for whom we are extremely thankful!

1) Alansari, Zaini. *Design, Simulation, and Testing of an Ultralight Radiator for Large-Scale Space-Based AI Data Centers.* Unpublished Manuscript, 2026.
2) Itawa et al. *Thermal Performance Evaluation of Space Radiator for Single-Phased Mechanically Pumped Fluid Loop.* Journal of Spacecraft and Rockets, February 2022

Simulation Results

Pareto Frontiers from the simulations below in Fig. 1.

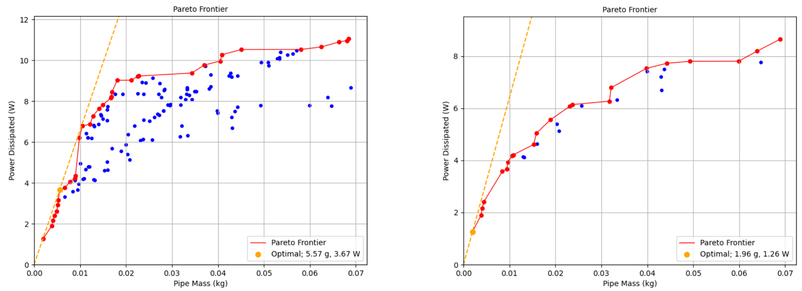


Fig. 1 — Pareto Frontiers showing data from all randomized runs of the simulation (left) and only runs of perfectly circular geometry (right).

- Orange lines show optimal mass-power ratio
- Both lines have near-equal slopes → two optimal geometries (Fig. 2)

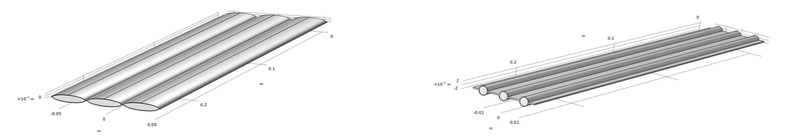


Fig. 2 — Optimal geometry found via Pareto Frontier (left) and the nearly-optimal circular pipe geometry (right), the latter of which is more feasible.

L:	$1.7cm$ semimajor axis	R:	$0.22cm$ radius
	$0.31cm$ semiminor axis		N/A (circle)
	$0.25cm$ spacer width		$0.64cm$ spacer width

Fabrication Methodology

Piping Network Fabrication

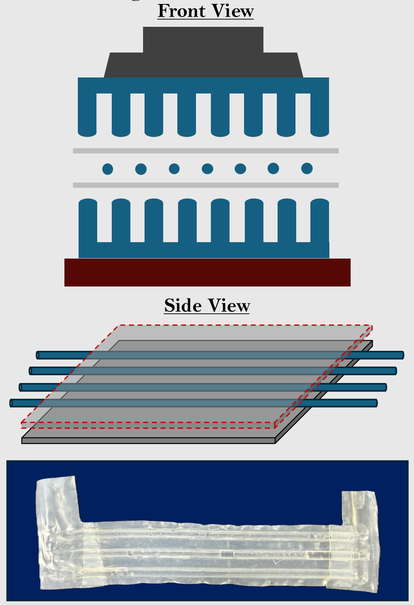
To fabricate the lightweight radiator, two sheets of ultra-high molecular weight polyethylene (UHWMP) are welded together using two heat sinks heated to $400°C$. The shape of the piping network is supported by placing perfluoroalkoxy alkane (PFA) cylindrical tubing between the heat sinks' ridges, which neither melts nor expands. This allows the PFA tubing to maintain its shape without disfigurement during welding.

Manifold Fabrication

During the piping network welding process, two non-welded UHMWP pieces are positioned at the end of the network. After the pipes are fabricated, these two strips are pressed using a flat aluminum block on all ends except for an inlet.

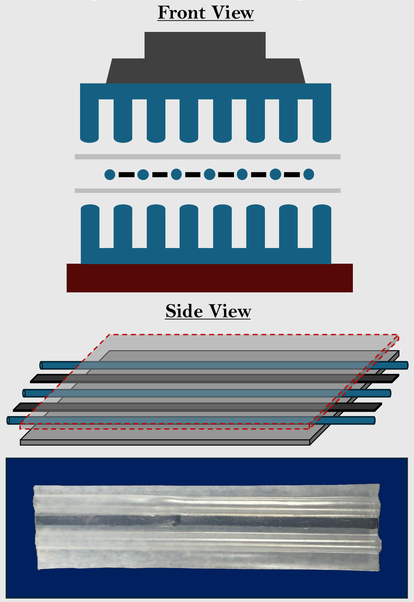
Fabrication Results

Diagrams of the welding process below in Fig. 3.



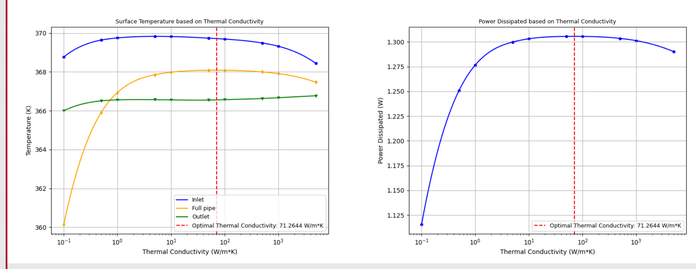
Future Fabrication

Diagrams of the graphite-spacer welding process below in Fig. 4.



Optimizing Thermal Conductivity

- Used COMSOL & optimal circular geometry
- Parametric sweep of thermal conductivity from 0.1 – $5000\ {}^{W}/_{m\cdot K}$
- Calculated the surface temperature of the pipe at the inlet (blue) and outlet (green), as well as the entire pipe-spacer surface (yellow), plotted against thermal conductivity (see below, left)
- Calculated power dissipated by the pipe, plotted against thermal conductivity (see below, right)
- Maximized power dissipated and found optimal thermal conductivity



Mechanisms

Heat spreads in two directions: flow-normal (radial) and flow-parallel (up- or downstream). At first, increasing thermal conductivity (k) improves flow-normal heat flow, improving heat rejection. However, for higher values of k, heat flows flow-parallel (i.e., across the pipe). As heat is better spread, the hotter regions with larger radiative contributions cool and radiate less heat away, causing a loss in efficiency of the radiator.

Future Work

- Simulate new geometries with pyrolytic carbon/graphite strips inside the welded spacers
- Look at power dissipation with graphite additions and possible benefits of anisotropic thermal conductivity
- Conduct another random-geometry simulation sweep with parameters: thickness and width of graphite, pipe length, and old parameters
- Create new fabrication processes to fit the optimal geometry